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Empowering robotic training with kinesthetic learning and digital twins in human–centric industrial systems

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ABSTRACT

This paper presents a human-centric mixed reality (MR) collaborative training platform that employs a kinesthetic learning technique in industrial robotic training, specifically focusing on robot pick–and–place (RPP) operations. Collaborating with ABB Robotics Vietnam, we conducted a user study to investigate the user experiences and practical perceptions of university students and novice trainees via the human–centric training assessment. The study compares the traditional training (TT) RPP classroom as a conventional method with a new collaborative MR RPP training approach (N = 50). The MR training features a digital twin (DT) of ABB GoFa™ CRB–15000 collaborative robot in an immersive 360° Digital–Objects–Based Augmented Training Environment (360–ATE) using Microsoft HoloLens devices. The research evaluated the impact of MR and DT on human–robot interaction and collaboration, user experience, task performance, knowledge retention, and interpretation, as well as differences in perceptions between the two novice cohorts under each training condition. The primary research question explores "Whether the MR collaborative training platform with DT integration in 360–ATE can serve as an alternative approach for novice students and industrial trainees in RPP operations?". The findings indicate that MR training is more engaging and effective in enhancing participant safety, confidence, and task performance, which also augments cognitive capabilities. The virtual contents on HoloLens, especially the DT, captured the attention and stimulated active learning abilities. Overall, participants in the MR cohort find the proposed training platform useful and easy to use. The platform has a positive influence on their intention to use similar 360–ATE–assisted training platforms in the future.

1. Introduction

The modern industrial systems and digital transition in digitalization rapidly evolve towards sustainability, resilience, and human–centricity, necessitating a comprehensive transdisciplinary approach [1–4]. This evolution emphasizes the critical role of human beings in the complex industrial systems designs and operations, integrating user–centric approaches, systems thinking, interdisciplinarity, and collaborative practices [5]. Digital tools and simulation techniques are key factors in this milestone transition, enabling the modeling of reality during the conception, training, and design stages [6]. These technological tools facilitate predicting system behaviors and human–system interactions, optimizing designs, technical training, and system maintenance to meet multidisciplinary requirements sustainably [7,8].

In this context, digital twins (DT) have gained prominence as a virtual representation known as a digital counterpart of a real–world object or system that allows the flowing, processing, and utilization of bi–directional communication data for various purposes such as simulation, integration, testing, monitoring, and maintenance [9]. Despite the inherent automation suggested by digitalization, human interaction remains indispensable [10,11]. Effective and friendly user interfaces are essential to cater to various roles and user groups within the industrial system, enhancing usability and societal impact. This collaborative interaction allows humans to learn and adapt to the systems over time, improving efficiency and effectiveness [12]. Methods such as model–based systems engineering, system dynamics, collaborative simulation training, and user experience design are crucial in

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developing these human–centric systems [13,14]. These approaches will aid the augmented cognitive capabilities, potentially supported by cloud–managed artificial intelligence and semantic technologies, further enhancing industrial systems and operations [15,16].

Immersive technologies (ImT), such as virtual reality (VR), augmented reality (AR), and mixed reality (MR), are useful practical platforms for virtualization, education, and training, particularly in an industrial setting [17–19]. They provide mixed physical–digital experiences and immersive collaborative interactivity on immersive devices. Thanks to these benefits, MR assists learners in learning/training through immersive and interactive solutions [20,21]. Multimedia training theory explains the benefits of immersive training via interactive multimedia digital objects [22]. Scoping reviews [6,23–26] discover favorable results for AR/MR–based training on boosting performance and efficacy of mixed reality–based training in manual assembly occupations with augmented cognitive capabilities. Traditionally, slideshows or demos are widely used in education and industrial training to offer procedural or operational process education. Prior works [27,28] employ collaborative immersive MR as a contemporary metaphor for transferring procedural information. Their results indicate improved performance and collaboration, but no significant differences or equivalences exist between the two conditions for higher learning outcomes. As digital technologies become more prevalent in classrooms and collaborative training practices [29–31], AR/MR application developers, educational practitioners, and even policymakers should better understand learners' AR/MR training technology usage experiences and opinions and their willingness to adopt them for learning.

This research is inspired to utilize a kinesthetic learning metacognitive approach [32], known as "learning by doing", for collaborative MR training in an industrial training scenario, namely the robot pick–and–place (RPP) operations. The research team collaborates with ABB Robotics Vietnam to develop the immersive 360° Digital–Objects–Based Augmented Training Environment (360–ATE) collaborative training platform and conduct a user study. This paper performs a comparative evaluation between the traditional training (TT) approach and the pilot collaborative mixed reality training approach for RPP operations with ABB GoFa™ CRB–15000 collaborative robot (cobot) station. With this MR training, a virtual DT of the cobot, user interface (UI), and human–robot interaction (HRI) based on the Microsoft MR HoloLens devices are constructed to undertake RPP operations (target instruction, robot path programming, and simulation). The study recruits two novice cohorts to reflect the typical groupings of ABB Robotics Vietnam trainees and RMIT University students with varied levels of technology proficiency without experience in RPP operations. The user study is developed to accommodate both cohorts' talents, experience, and time commitment. Overall, the paper reports an empirical comparison of proposed MR training approach to the conventional TT with knowledge retention, knowledge interpretation, and task performance measures on the final practical assessment at cobot station under ABB specialists' supervision.

Building upon the provided background, motivation, and our previous works [33,34], the present study aims to explore how a human–centric collaborative mixed reality training platform can be integrated into both industry and education. It also examines students' understanding of technical works to enhance their safe practical experience and augmented cognitive capabilities. The overarching research question is "Whether the MR collaborative training platform with DT integration in 360–ATE can serve as an alternative approach for novice students and industrial trainees in RPP operations?". To address the research gap, this work explores novice users' subjective impressions of the collaborative MR RPP training platform, specifically the user's psychometric perceptions and intentions [35,36] to use a similar human–centric industrial training platform. The study considers various potential elements that might be operating while determining whether or not to embrace the MR collaborative training approach. Compared to previous works and practical studies, the study provides the actual training through a relationship with a robotics and automation organization invoking a validated hands–on technical examination. In summary, we aim to give a thorough knowledge of user evaluation and how individual perspectives affect user acceptance with this collaborative MR training platform. In this experimental research, the potential contributions are listed as follows:

1. Introduce a mixed reality collaborative training platform with kinesthetic learning strategies and digital twin integration in robot pick–and–place operations
2. Compare the proposed immersive human–centric training approach with the traditional training method, then highlight the impact on user experience and training outcomes of the 360–ATE training platform
3. Present empirical study as evidence–based quantitative findings to researchers, developers, educators, policymakers, and stakeholders to enhance the deployment of these technologies in education and industrial settings

For the paper content summary, Section 1 has developed the aims and motivation for this article. This is followed by a literature review of immersive training, human–robot interaction and collaboration, user validation, and usage of digital twins in the industry in Section 2. Section 3 then explains the research methodology and user experience design. The analysis results of the user study are given in Section 4. Discussion of the findings and limitations are presented in Section 5 and finally, Section 6 provides the conclusions drawn from this study.

2. Background and related work

The integration of human–centric immersive training technologies is believed as crucial in terms of enhancing the effectiveness of training and education. This approach not only allows a more engaging learning environment but also facilitates human–machine interaction and collaboration for optimizing workflows and improving productivity. Since organizations increasingly adopt advanced training methods, user experience validation gains more prominence to ensure that these technologies meet the needs and expectations of learners. Furthermore, the utilization of digital twins in training offers a revolutionary way to simulate real–world scenarios, providing learners with hands–on experience in a risk–free setting. Besides these advancements, there remains a significant research gap that needs to be addressed to fully harness the potential and enhance their deployment in various sectors. The following presents each of these critical areas, highlighting the interconnections, research gaps, and challenges for future research.

2.1. Immersive human–centric training technologies

Immersive training has gained significant traction in recent years due to its potential to simulate real–world environments and scenarios [37,38]. These technologies enable users to engage in highly interactive and realistic experiences, enhancing learning outcomes compared to traditional methods [39,40]. Various sectors, including healthcare [41], military [42], and aviation [43], have adopted immersive training to improve skill acquisition, retention, and performance.

VR offers a fully immersive environment, isolating users from the real world and placing them in a simulated space where they can practice tasks without real–world consequences. For instance, medical trainees can perform virtual surgeries, allowing them to make mistakes and learn without risking patient safety [35]. AR, on the other hand, overlays digital information in the real world, enhancing the user's perception and interaction with their environment. This is particularly useful in maintenance and repair, where technicians can receive real–time, step–by–step guidance while working on complex machinery [44]. MR combines elements of both VR and AR, creating a hybrid environment where physical and digital objects coexist and interact in real–time. This can provide even more nuanced training scenarios, like collaborative tasks requiring both real and virtual components [45,46].

Despite the advances and benefits of immersive technologies in 360–ATE, there is a notable gap in the research concerning collaborative training using these technologies. Most existing studies [47] focus on individual training, evaluating the effectiveness of VR, AR, and MR on a single user's performance and learning. However, many real–world tasks require teamwork and collaboration, critical skills in sectors such as emergency response, healthcare, military, and industrial operations [48]. Collaborative training in immersive environments [49,50] could help trainees develop not only individual skills but also the ability to work effectively as a team under various scenarios.

2.2. Digital twins in training and education

Digital twins provide real–time data and analytics, which create dynamic simulations that mirror real–world processes [10,51–54]. The integration of digital twins with immersive technologies has introduced a promising avenue for enhancing collaborative training [55]. With this combination, DT offers a unique and interactive training environment, improving learning outcomes and operational efficiency. Besides, DT have been trendy and applied in industries such as manufacturing, healthcare, and aerospace for predictive maintenance, performance monitoring, and optimization [56–58]. In training contexts, they enable the creation of realistic scenarios where trainees interact with accurate models of complex systems [59]. For instance, DT allow workers to train with virtual models of machinery, and understand how to operate and troubleshoot without physical risks or downtime [60,61].

Despite the clear advantages [54], there is a notable research gap in the use of digital twins in immersive collaborative training [62,63]. Most existing research and applications focus on the digital transition and digitalization of industrial systems and do not focus on the utilization of DT in human–centric training, personal skill development, and human resource development at the industrial level. Previous studies have explored and validated the use of digital twins for collaborative training, with a focus on system performance and human safety [64–67]. Thus, there is a lack of research on how humans perform, interact with, and perceive these systems as well as methods to measure the human factors in this context [9].

2.3. Human–robot interaction and collaboration in industrial settings

The rise of industrialization has brought robots into automated manufacturing in which their role is rapidly evolving from caged robots, it is stepping into the world of human–robot collaboration (HRC), where robots share workspace with humans [61,68–70]. This paves the way for a unique yet productive combination: human adaptability and problem–solving working alongside robotic system precision and tireless execution. Effective interaction is crucial for this collaboration, requiring specialized training for the operators [71].

The term industrial settings represents environments such as factories, manufacturing plants, and production facilities [72]. Key characteristics of industrial settings include the integration of heavy machinery, streamlined production lines, and meticulously designed operational workflows, all aimed at efficiently transforming raw materials into finished products [73]. In addition, these environments are engineered to withstand demanding conditions and adhere to rigorous safety protocols, ensuring the health and well–being of the workforce. This understanding of industrial settings is essential as we investigate the implications and applications of immersive technologies within these critical operational domains, especially the workforce training.

A driving force behind HRC is the emergence of collaborative robots or cobots for short. These smaller robots are specifically designed to interact safely with humans [61,74,75]. HRC promises to revolutionize workplaces by capitalizing on the unique strengths of both human operators and advanced robotic systems. Robots excel at repetitive tasks, freeing humans to focus on their areas of expertise–dexterity, adaptability, and problem–solving [76]. This harmonious collaboration has the potential to boost productivity, improve product quality, and allow humans to tackle more high–level tasks. However, safety remains the foundation of successful HRC [77]. Studies have highlighted humans' challenges in understanding and communicating with robotic systems [78]. It emphasizes the need for clear communication protocols, user–friendly robot interfaces, and robust safety features. Moreover, effective collaboration does not only require technical aspects but also building trust between humans and robots. Research by Zacharaki et al. [79] explores the psychological aspects of human–robot interaction. Although the cobot is designed for close interaction or requires human guidance during operation, training programs are necessary to emphasize these protocols and equip workers to identify potential hazards and understand robot limitations [80,81]. Collectively, HRI training fosters open communication and familiarizes workers with robot capabilities.

2.4. User experience validation in training and education

Human factors are primary essential considerations for human–centric design and augmented cognitive capabilities in industrial transition. Consequently, it is crucial to provide comprehensive training for operators of industrial systems [73]. Well–trained workers can operate these systems with maximizing efficiency while minimizing the risk of encountering issues. In addition, requiring troubleshooting or maintenance, experienced technicians can complete tasks more swiftly, thereby reducing system downtime [82].

Presently, the most common training methods are on–the–job training and traditional classroom style [83,84]. These methods involve employees learning tasks and skills via theories training and performing them directly within the work environment [85,86]. They allow trainees to gain practical experience under real conditions, often with guidance from experienced colleagues, enhancing both their competency and confidence. However, these methods disrupt normal workflow, as new workers need time to become familiar with the system. Moreover, it increases the risk of human error during the initial phase of training leading to unsafe conditions, especially for novices. Alternatively, another training approach is with a simulation tool [6,87–89]. It involves acquiring and improving various skills through the use of digital models capable of replicating a different situation and processes from real scenarios [90]. This approach is optimal for uncompleted installed systems or requires updates, and enhancing user safety.

The incorporation of human–centric simulation training, especially immersive technologies like VR, AR, and MR, holds promise for enhancing training and education standards within industrial and engineering contexts [91,92]. However, it is necessary to conduct more research to investigate the user experience and evaluate the practicality and efficiency of these alternative simulation–based methods beyond the technical performance aspects.

2.5. Summary of research gaps

Immersive technologies have been introduced to enhance individual training outcomes across various sectors. Most existing studies focus on personal performance, overlooking the importance of teamwork and social dynamics essential in transdisciplinary environments [93–96]. The integration of immersive technologies with digital twins presents both opportunities and challenges in training and education [97,98], but there is still limited research on their effectiveness in industrial collaborative settings [17,99,100]. A significant gap exists in the integration of DT with ImT, as many current implementations focus on standalone immersive experiences without fully leveraging the real–time data and simulation capabilities that digital twins can provide [101]. Although there are various applications of ImT and DT in architecture and construction [102–104], there is still limited research and comparisons in modern practices in terms of integrating digital twins into education and training [54,105]. This lack of integration may hinder the effectiveness of training programs, as they often fail to replicate the complexity of real–world environments and processes [106]. By addressing these gaps, we can enhance the quality and effectiveness of training and education in industrial settings, ultimately leading to better outcomes for learners and organizations alike [105].

This research gap extends to the application of digital twins in training, especially in industrial settings. Despite the potential to create realistic human–centric and interactive training scenarios, it has primarily been explored for individuals rather than collaborative training. Addressing this gap requires investigating how these technologies can facilitate effective cross-team interactions, communication, problem-solving, and decision–making in a shared augmented space, enhancing individual and team–based skills in industrial contexts. Overall, this work emphasizes a critical gap in integrating mixed reality technology and digital twins for kinesthetic collaborative training in industrial human–centric applications.

3. Experimental methodology

The study applies two training conditions in RPP operations. The participants are randomly divided into two equal cohorts to attend these two training conditions, traditional classroom or immersive mixed reality platform. To make the experiment more manageable, participants are free to choose their suitable condition training section schedule based on the random selection results as has been mentioned. In the first condition, participants take part in a traditional RPP classroom training as usual at industrial partner R&D centers and experiment with a brand new ABB GoFa™ cobot. On the other condition, participants use the pilot MR RPP training solution with MR HoloLens devices for training, simulating, interacting, and operating with the DT cobot station. Afterward, both novice cohorts will be examined with the same knowledge retention questionnaire and experiment on the physical cobot station for practical tests.

3.1. Research questions and hypotheses

The main research question is *"Whether the mixed reality collaborative training platform with digital twins integration in 360–ATE can serve as an alternative approach for novice students and industrial trainees in robot pick–and–place operations?"*. To investigate the benefits, challenges, and potentials of the human–centric MR training platform, this study will address the following research questions (RQs) and hypotheses (Hs). The combination of findings from these RQs is used to clarify as evidence for the three main contributions of this study as mentioned in the Introduction (Section 1).

- **RQ1.** What are the differences between the two cohorts in terms of knowledge retention and interpretation?
→ **H1.** The knowledge retention and interpretation of the two training cohorts (TT and MR) are the same.

- **RQ2.** What are the differences between the two cohorts in terms of user evaluation of engagement and knowledge delivery as a learning/training platform?
→ **H2.** The levels of engagement with the training content of the MR cohort are higher than the TT counterpart.

- **RQ3.** Does the integration of digital twin cobot attract the user's attention, aid the simulation training process, and ensure safety rather than complicate the training process?
→ **H3.** The usage of digital twin cobot enhances safety and aids the simulation training process.

- **RQ4.** What are the different efforts between the two cohorts in training assessments of the industrial partner?
→ **H4.** The MR cohort spends less effort than the TT counterpart to complete the training assessments.

- **RQ5.** Can the incorporation of mixed reality and digital twins in collaborative human–centric training for robot pick–and–place operations contribute to enhancing technical competence and augmented cognitive capabilities among students and trainees in automation, and extend to related training fields?
→ **H5.** There is a rejection of using the proposed platform for related training of its technology adaptability and knowledge requirement.

3.2. Research ethics

All participants signed a consent form before participating in the study. All procedures conformed to the National Statement on Ethical Conduct in Human Research (NHMRC,2007). The study was approved by the College Human Ethics Advisory Network (CHEAN) at RMIT University with project number **#26910**. Privacy and security are significant concerns with personal data collection, especially participant demographic data. To preserve the anonymity of our participants, the analysis data does not include personal data and is non–identifiable or re–identifiable.

3.3. Development of training program

The training program was developed over a year with ABB Robotics Vietnam, at ABB Robotics & Automation Solution Center – RASC, Ho Chi Minh City, and ABB Robotics Technical and Service Center – RTSC, Bac Ninh (since 2022). The program establishes two training conditions: the traditional training classroom and the pilot MR training platform on RPP operations. To ensure the safety of participants, the research team and industry partner (ABB Robotics Vietnam) decided to use a brand–new ABB GoFa™ CRB–15000 cobot due to its safety protocol and user–friendly features that offer collaborative operations with side–by–side training.

3.3.1. Traditional training classroom

Robot pick–and–place operations, or RPP operations, are essential training when working with robotic manipulators in automation. Initially, there are consultations with the specialist and engineering team from ABB Robotics Vietnam to extract a standard RPP operations training from the whole professional training program. Discussions have carried out the final arrangements on the standard RPP training program used in this study, completely instructed by ABB specialists. Traditionally, the current training approach uses slides, photos, demo videos of RPP operations, and a practical experiment with the ABB GoFa™ cobot station. In detail, the specialist team prepares a classroom–style presentation that introduces the standard cobot cell, cobot programming techniques, operating environments, and detailed instructions for the RPP operations.

3.3.2. Collaborative mixed reality training platform

Development and system architecture. The MR training platform is developed based on the material resources, database, and demands of the industrial and academic partners. The proposed training system architecture is made up of three essential components including an ABB GoFa™ CRB–15000 cobot station, a central broker, and immersive user interface devices (HoloLens 1/2) based on our previous works at ABB Robotics Vietnam [33,34]. Each component details a structure and how they interact to achieve the collaborative function of the MR training platform as Fig. 1. The platform development is divided into three main stages: data collection, modeling, and programming.

In the data collection stage, essential information is collected, including RPP descriptive procedures, simulation, digital industrial environment setup, equipment, and tools. Since MR developers do not know about engineering and robotics, it is a challenge for us to understand the details of RPP procedures. In the modeling stage, a digital twin cobot is mirrored on the HoloControl application, which is optionally

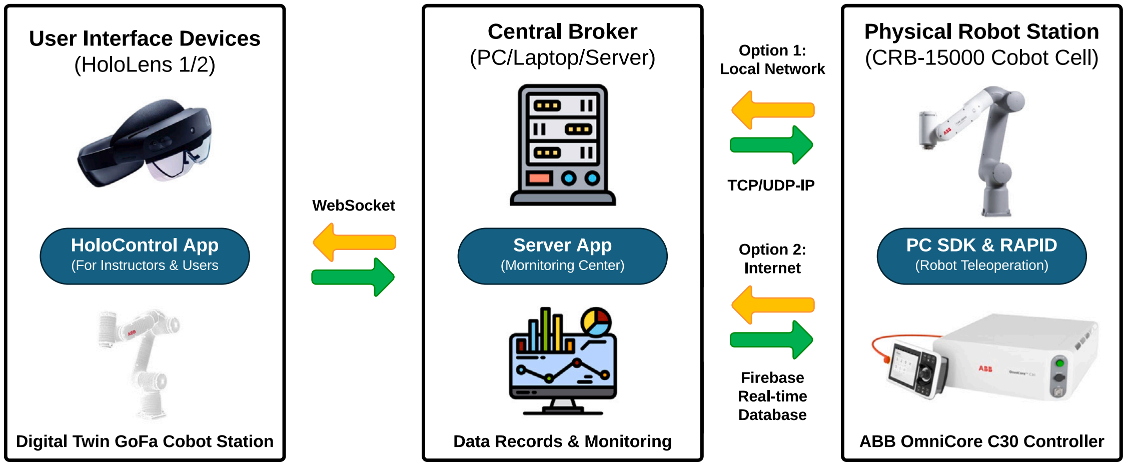

Fig. 1. The collaborative mixed reality (MR) training platform architecture and transmitting/reception connections layout between the HoloLens Devices and ABB GoFa™ cobot.

linked to the ABB GoFa™ cobot station at ABB R&D centers or operates independently as a digital holographic cobot station at any location. With the programming state, the research team must develop two different applications. One is a collaborative holographic application on HoloLens devices for instructors and users, and the other is a server application, that allows the connection between the DT station and the physical one and runs on a laptop, PC, or physical/cloud server as a central broker.

For data transmission and reception, the immersive user interface devices utilize WebSocket which establishes the communication between UWP devices (HoloLens 1 and 2) and the Central Broker (PC, Laptop, or Server). There are two communication protocols between the Cobot Station and Central Broker: one is TCP/UDP-IP in the private local network, and the other is real–time database services on Firebase. The data recording and monitoring are supervised and stored on the "Central Broker" device. Due to the IP and cyber security policy of ABB Group, the real–time communication between the digital twin station on UI devices and the physical cobot station is customized by establishing private networking following the industrial standard for this training system. The technical details of all hardware devices (including external devices) used in this study are presented in Table 10.

Digital twin and user interface. The research team develops the digital twin based on the industrial partner's mega 3D CAD resources. The MR training platform for the RPP operations is created using the Unity game engine with official support MRTK 3 packages. The model is first optimized in SolidWorks at a low level of detail to maximize the rendering time consumption on the MR devices. Then it is imported to Blender for 3D model joint reconstruction as in Fig. 2a, and exported as a stereolithography file (STL–Standard Triangle Language format) for Unity. The final STL format game object is imported to Unity for scripting combined with network communication to establish the digital twin on 360-ATE with the real–time bi–directional data feedback to the physical ABB GoFa™ cobot station as in Fig. 2b.

The DT cobot station could be anchored to a surrounding environment and the physical cobot base for the 360-ATE manually or auto–configuration with a QR code. The training platform supports up to 10 participants remotely connected to the same PC server application (only two active users at the time – one trainer and one trainee, others are viewers). When a user's flat hand is placed inside the range of the four deep tracking cameras, the Main Menu will pop up and present the main control panel at the start screen. This interactive menu, shown in Fig. 3a, gives the user comprehensive access to all the training features, including the event log, cobot control center, and server connection hub (option – connect to physical cobot station for practical training). Besides the main configuration menus, the system allows the user to monitor and interact with the cobot during operations. All the cobot status and data will be displayed on the Server Menu, Control Hub, and Event Log when the error appears. The virtual objects could be anchored anywhere in the physical environment by manual position using hand gestures.

This training condition implements kinesthetic learning by providing safe closed realistic interactions with the digital twin cobot, cobot targets, and a programming review portal with the support of a collision avoidance feature on the HoloControl app. The collision avoidance allows the users to safely program the physical cobot by interacting with the digital twin via the support of virtual mesh as Fig. 3b.

3.4. Participants

A group of 89 participants was recruited from RMIT University and ABB Robotics Vietnam. The recruitment process is designed to encompass a broad spectrum of individuals with varying levels of expertise in the field of technology. While students from all academic programs are initially considered, participants possessing professional robotics and technical backgrounds are subsequently excluded. The participants represent diverse technical backgrounds, including electrical/electronic engineering, mechatronics, and robotics. After selection procedures and final confirmation, 39 individuals withdraw from the study. A final sample of 50 participants (56.18% of the original population) includes 29 university students and 21 novice trainees.

This approach enables the generalizability of research findings to a broader demographic, comprising 58% of the STEM students and 42% of the industrial trainees, both of whom are novices in RPP operations, with a gender distribution of 92% males and 8% females. Mathematically, the participant's gender ratio above is also equivalent to the reports about robotics engineer demographics and statistics in the US (93% males and 7% females) [107]. Within this final participant pool, they are randomly assigned into two equal training cohorts (25 individuals per cohort).

3.5. Research procedures

For both cohorts, each participant receives one training condition in robot pick–and–place operations, under the instruction of the industrial specialist team. The study is conducted on–site at ABB RASC – Ho Chi Minh City and ABB RTSC – Bac Ninh. The participants arrive for each condition training session at ABB R&D centers (4 sessions in total, 2 per condition). The total training time of the two conditions is designed to be approximately the same (one hour), excluding the hardware setup, preparation, survey, assessments, and break time. The total time for participants to complete this study is approximately 2.5

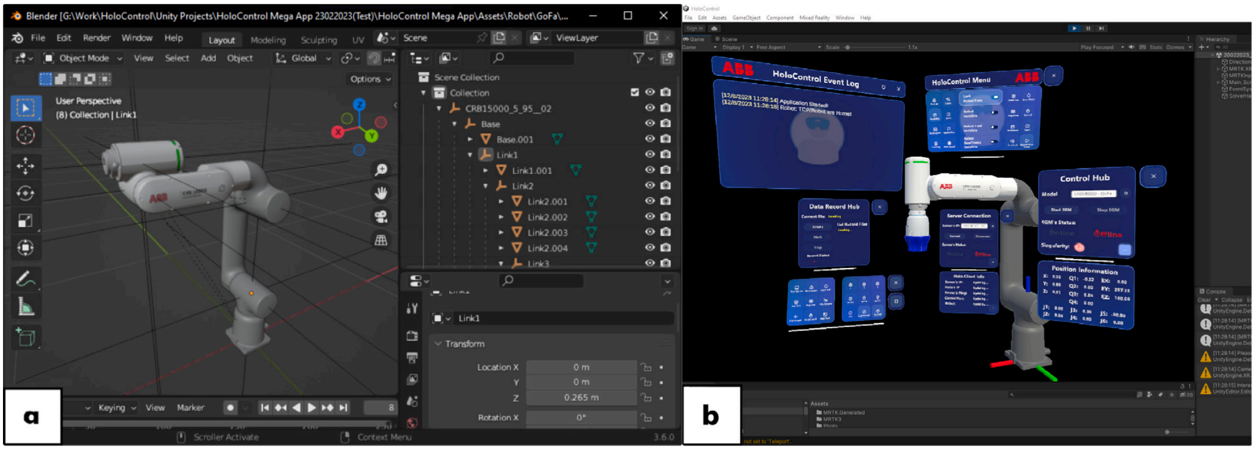

Fig. 2. (a) Reconstructed 3D model of ABB GoFa™ Cobot in Blender. (b) View of its digital twin and training interactive interface in Unity.

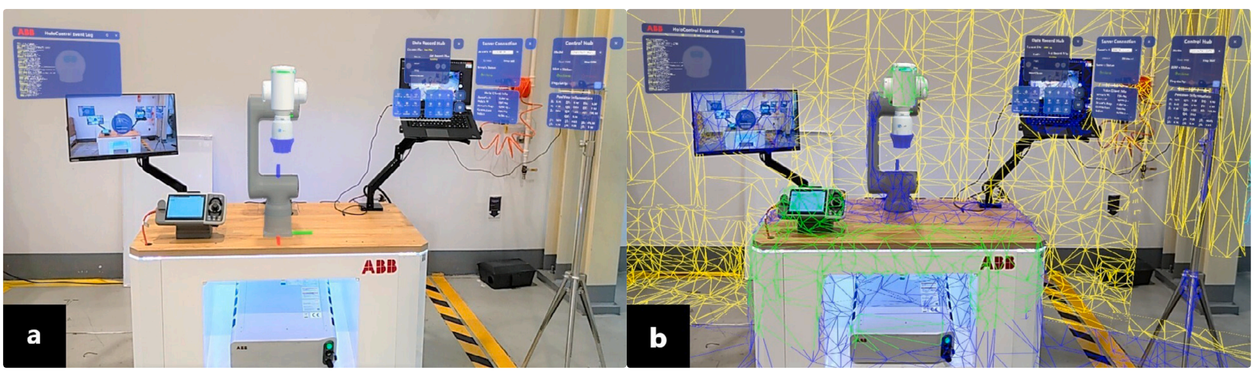

Fig. 3. (a) The digital twin cobot station overlays the physical one in human–robot interaction training. (b) An augmented view of human–centric collaborative robot pick–and–place (RPP) training application on HoloLens devices (digital aid mesh for collision detection feature).

Table 1
The summary of the research time frame between two training conditions (TT & MR).

No.	Training procedure	Traditional training (TT)	MR training (MR)	Time frame
1	**Welcome & Pre–Surveys**	Individual responses	Individual responses	
1.1	Demographic Survey	Online form	Online form	
1.2	Digital Consent Form	Digital signature	Digital signature	Total ~15 min/participant
1.3	PIL and VARK Surveys	Online survey	Online survey	
2	**HSE/SA Training**	Section group training	Section group training	Total ~20 min/group
Break (~10 min)				
3	**Robot Pick–and–Place Operations Training**	Classroom style and on–site training	On–device training and live instruction streaming from the instructor's MR headset	Total ~30 min/participant
3.1	Theoretical Training	TT cohort training with PowerPoint slides	MR cohort training with live instruction on–screen	~10 min/cohort
3.2	Training Station	Physical cobot cell	Virtual cobot cell (not DT)	~10 min/participant
3.3	Practice Section	With the physical cobot cell	With a digital twin cobot on MR devices side–by–side with the physical cobot	~10 min/participant
Break (~10 min)				
4	**Training Assessment**	Individual assessment	Individual assessment	Total ~40 min/participant
4.1	Knowledge Retention	Online quiz	Online quiz	~10 min/participant
4.2	Knowledge Interpretation	With physical cobot cell	With digital twin side by side the physical cobot cell	~30 min/participant
4.3	NASA–TLX Assessment	Individual response with assistance	Individual response with assistance	
Break (~10 min)				
5	**Post–VARK Surveys**	Online survey	Online survey	Total ~05 min
6	**Post Surveys**	Not applicable	Individual Response	
6.1	TUX	–	Online Survey	
6.2	SDS	–	Online Survey	Total ~30 min
6.3	SCLS	–	Online Survey	
6.4	EPQ	–	Online Survey	

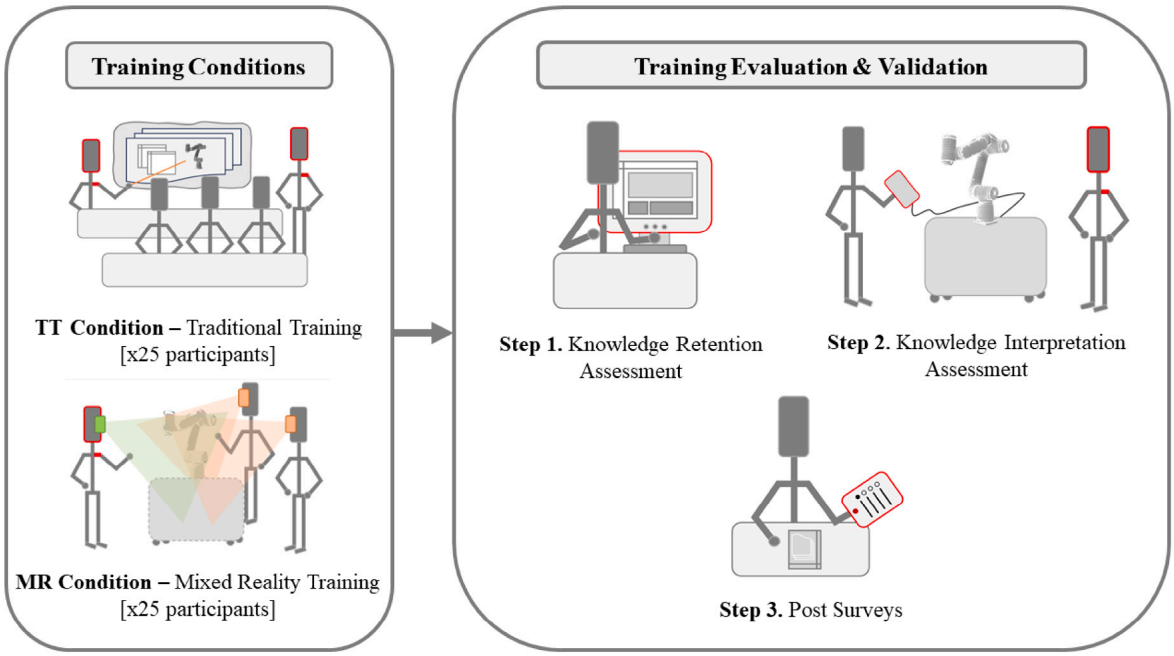

Fig. 4. Research procedures for robot pick–and–place (RPP) operations training program on both traditional training (TT) and mixed reality (MR) training conditions.

hours (140 minutes for each TT participant and 170 minutes for each MR counterpart), a detailed summary is presented in Table 1.

Upon arrival, the participants sign in as visitors and are provided plain language statements and a participant consent form (as approved by RMIT University College Human Ethics Advisory Network – CHEAN). Each participant completes the onboarding questionnaires, including demographics, personal innovativeness level, and VR/AR/MR prior experience before stepping into the Health–Safety–Environment–Security and Sustainability (HSE/SA) training delivered by ABB HSE specialist staff. After the safety training, the participants follow the research procedures as Fig. 4. The TT cohort attends the traditional classroom–style training in the office while the MR one joins the 360–ATE at a different location as Fig. 5. At the end of each training session, participants have a practical experiment with the DT cobot and/or physical ABB GoFa™ cobot station to practice what was trained.

3.6. Data collection and analysis

The data collection involves observations during the experimental procedures, as well as the administration of pre–test and post–training test questionnaires as Fig. 6. Additionally, there are several tests/surveys related to technical and user experience that the participants need to pass during the study. These questionnaires are built using Google Forms, and a QR code is provided for participants to complete on their mobile phones or provided tablets.

- **Consent Form and Demographic Survey.** Before the HSE/SA training, participants are asked for their understanding, agreements, and signatures. An online demographic survey is provided to collect personal information (name, age, gender, etc.), personal innovativeness level (PIL) [108], and prior VR/AR/MR experience. These databases will help in the summary of the study group and final analysis.

- **VARK Questionnaire.** After the HSE/SA training, participants are invited to complete a pre–VARK learning style questionnaire with 16 questions [109]. The VARK questionnaire examines perceptual preference for several ways of knowledge transmission/learning styles [110] within four modalities, including Visual (charts, graphs, or symbolic devices – Vs), Aural/Auditory (oral lessons and conversing – As), Read/Write (written material – Rs), and Kinesthetic (learning by doing – Ks). This post–VARK would be asked again after the technical and practical tests.

- **Knowledge Retention (KR) and Knowledge Interpretation (KI).** After each training condition, the participant is asked to take the human–centric training assessment, including the knowledge retention test, and knowledge interpretation test. The KR test is written using a multiple–choice format with 8 questions, designed to evaluate how much factual knowledge is retained from both training conditions [111] (KR pass score > 5/8). The KI test verifies the performance of basic robot pick–and–place operations at the physical ABB GoFa™ cobot station (KI pass score > 79/100). These two tests are under the instruction and supervision of the ABB specialist team based on standard training verification procedures.

- **NASA–TLX and Training Engagement (TE) Survey.** To investigate the workload on the RPP operations, the NASA–TLX assessment, known as the NASA Task Load Index, is invoked after the human–centric training assessment (KR and KI tests). This assessment categorizes the amount of demand or workload in six areas: mental (MD), physical (PD), temporal (TD), performance (P), effort (E), and frustration (F) [112]. Participants are invited to perform a weighted version of the NASA–TLX [113,114] in their RPP operations to assess the impact of cognitive strain generated by the consequence of each kind of training condition. In addition, the training engagement criteria include difficulty, efficiency, satisfaction, and task completion. This TE survey examines the engagement level of the participant on their training condition (based on the Likert scale of 1–5, TE > 3 indicated high training engagement).

- **MR cohort surveys.** At the end of the study, the MR cohort is invited to complete four more surveys related to their training condition, including training user experience (TUX) [35,115], simulation design scale (SDS), satisfaction and self–confidence in learning scale (SCLS), and educational practices questionnaire (EPQ) [116]. Participants were given statements and utilized a

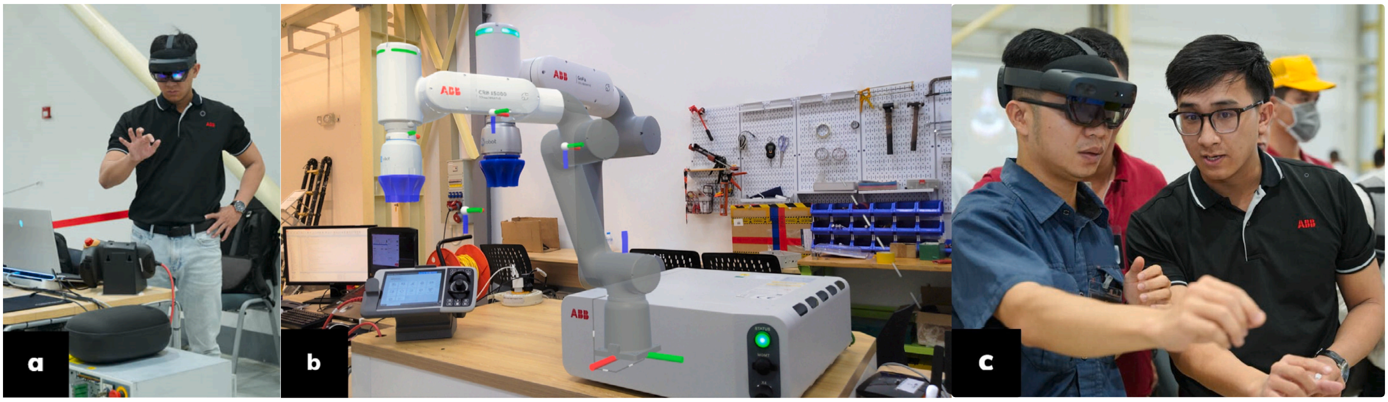

Fig. 5. (a) ABB specialist staff sets up the mixed reality (MR) robot piick–and–place (RPP) training station. (b) Collaboration ABB trainer's and trainee's views on HoloLens devices in MR RPP training, see–through with the digital twin ABB GoFa™ CRB–15000 cobot station overlays the physical one. (c) Trainer (right) is carrying the robot pick–and-place training instruction while the trainee (left) observes the operation task.

5–point Likert scale to indicate how much they agreed with each statement (strongly [dis]agree, [dis]agree, and neutral).

To ensure the integrity and validity of this empirical study, several key factors are taken into consideration to minimize user bias. First, it is essential to ensure a diverse participant selection that represents various demographics, including age, gender, and professional background, capturing a wide range of perspectives and experiences. Implementing randomization of participants to different conditions/training groups further reduces the influence of confounding variables, ensuring that results are not systematically skewed. Across the experiment, a standardized protocol is utilized for data collection, including clear instructions and consistent environments to eliminate variations that could affect user experiences.

Additionally, assuring the anonymity and confidentiality of participants encourages honest feedback, reducing social desirability bias. Conducting pilot tests of study instruments allows the research team to identify and address potential biases in questions or processes before the main study begins. Implementing mechanisms for continuous feedback during the study can help identify and mitigate any biases that arise in rea–time, allowing for adaptive modifications to the study design. Finally, evaluating and analyzing any non–responses or dropouts is crucial to assess if certain groups were underrepresented, as this could skew the results. Incorporating these mentioned factors into the user experience reporting process can better ensure the validity of the findings and minimize potential biases that could affect the outcomes of this study.

The data analysis in this study aims to examine the impact of integrating MR technology into industrial training to enhance robotic experiences, specifically RPP operations, compared to the traditional training technique. Initially, the Shapiro–Wilk test is employed to check the normality of collected datasets. Since all datasets are not a normal distribution, non–parametric statistics, specifically the Mann–Whitney U test and Wilcoxon's signed–rank test, are invoked to compare human–centric training assessment outcomes between cohorts, NASA–TLX, training engagement, and the impact of training conditions on participants' preferred learning style. Furthermore, Spearman's rank correlation test and Cronbach's Alpha are also utilized to examine the correlation between KI and KR, and the internal consistency of survey items on TUX, SDS, SCLS, and EPQ.

Mathematically, we perform statistical comparison tests for quantitative results and thematic analysis on the dataset with independent and dependent variables for the research findings. The analysis research members are assigned to conduct thematic preliminary analysis using Excel (for general analysis), Matlab (for virtualization), and RStudio (data analysis and testing), and then the results and findings are verified, detail analyzed, and virtualized by the research team, especially the Principal/Co–Investigators, and ABB training specialist team.

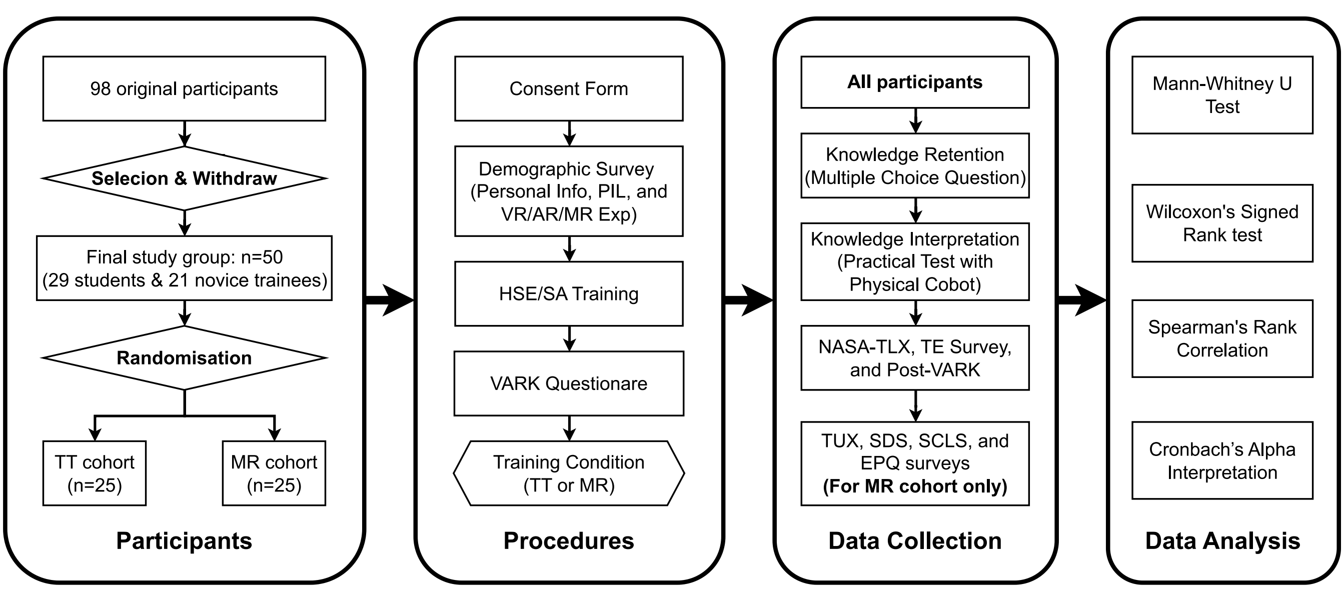

Fig. 6. Flow diagram of user study design.

Table 2
Summary of the participants' characteristics.

Variable Gender	Students n(%)	Trainees n(%)	N n(%)	None Exp n(%)	VR/AR/MR Exp n(%)	PIL
Male	26 (89.7%)	20 (95.2%)	46 (92%)	31 (93.9%)	15 (88.2%)	4.00
Female	3 (10.3%)	1 (4.8%)	4 (8%)	2 (6.1%)	2 (11.8%)	4.02
Total	**29 (58%)**	**21 (42%)**	**50 (100%)**	**33 (66%)**	**17 (34%)**	**4.02/5.00**

Table 3
Spearman's rank correlation between KR–KI results & the pass rates on two cohorts.

Training	KR & KI correlation		Pass rate (PR)	
Cohort	r_s	ρ (2–tailed)	KR	KI
MR	0.4123	0.0406[a]	1.00	0.92
TT	0.4699	0.0178[a]	0.76	0.52

[a] $\rho < 0.05$.

4. Experimental results and analysis

4.1. Participant demographics

The effectiveness of the proposed methodology is demonstrated through an experimental approach. Data collection involves the use of multiple methods, including assessments, questionnaires, and surveys. The author conducts experiments to evaluate the influence of integrating the MR platform on RPP operations. A total of 50 individuals (29 university students – 21 novice trainees, 92% males – 8% females) participates in the study. The participants are divided into two equal cohorts for two training conditions (MR or TT), as presented in Table 2.

The average age of participants in the final group is 22.8 years (standard deviation – SD is 1.85), ranging from 20 to 28 years. Participants' personal innovativeness level (PIL) exhibited an average score of 4.02, spanning from level 2 to 5. Regarding prior immersive technologies experience, 66% of participants have no experience, while 34% have used VR/AR/MR technology before (including 14/17 individuals – 82.4% with VR, 12/17 – 70.6% with AR, and 3/17 – 17.6% with MR, no one has professional experience with these immersive technologies) as Table 2. In addition, most of the participants prefer the kinesthetic learning style as presented in Table 5.

4.2. Analysis of human–centric training assessment

The Spearman's rank correlation test is employed to examine the correlations between the knowledge retention (KR) results and knowledge interpretation (KI) in both cohorts. The correlation results are presented in Table 3, illustrating the correlations between the technical and practical outcomes of the two cohorts. The findings in both cohorts reveal a significant positive correlation between KR and KI (MR – $\rho = 0.0406$, TT – $\rho = 0.0178$, $\rho < 0.05$). Moreover, the KR and KI test pass rates are also presented in Table 3. Success in the KR test does not guarantee that the participant passes the KI test which is strongly indicated in the TT cohort (MR–KI PR = 92%, TT–KI PR = 52%).

Furthermore, the Mann–Whitney U test is conducted to examine the RPP operations competence after the training between cohorts. The results of the Mann–Whitney U test indicate significant differences observed between the two cohorts in KR and KI tests as shown in Table 4. Both training conditions finally provide significantly distinct levels of factual knowledge and performance. Consequently, it concludes that MR training is considered a more successful and efficient platform in RPP operations, as indicated in the correlation graph Fig. 7.

Regarding the KI test, only 52% of TT participants pass the test in the limited–time performance while only two MR participants fail with a 70/100 score (leading to a total pass rate of 92%). Moreover, the KR results have a medium gap from both cohorts since the MR–KR pass rate is 100% with a single training session. Hence the MR cohort could remember the key points via the kinesthetic training. In comparison, the TT cohort maintains a 76% KR pass rate. Besides that, the analysis finds a correlation between high KR and KI scores in the MR cohort compared to the TT one. The correlation shows that participants who were better in the retention test were also better in the interpretation test. In contrast, the TT participants who pass the KR test are not guaranteed success on the KI test.

4.3. Analysis of NASA–TLX and training engagement survey

As presented in Table 4, there is no significant difference between the two cohorts in the NASA–TLX. However, the performance demand in NASA-TLX logically observes significant differences ($\rho = 0.0031$, $\rho < 0.05$), which provides additional evidence that MR training (Mean = 89.00, SD = 9.46) and has a positive impact on the training performance compared to TT one (Mean = 72.40, SD = 22.5, KI pass score is 80+). In addition, Fig. 8 demonstrates that the MR cohort performance is more consistent than TT. The other five factors are no different indications via the Mann–Whitney U test. However, it is observed that all participants reported mental, temporal, and effort demands on the NASA–TLX scale with mean scores around 40–50/100 as shown in Table 4, exhibiting a wide data range. The statistical plots of the NASA–TLX data for both cohorts are provided in Fig. 8.

4.4. Analysis of participants' preferred learning style

Wilcoxon's signed–rank test is employed to compare the differences in participants' VARK preferred learning style of both cohorts before and after attending their training conditions. The results of Wilcoxon's signed–rank test reveal a significant change in the kinesthetic modality (Ks) in the pre–VARK questionnaire and post–VARK questionnaire in Table 6 (MR–Ks $\rho = 0.019$, TT–Ks $\rho = 0.044$, $\rho < 0.05$). In terms of Ks on post–VARK results, the MR cohort has 24 participants with a positive sign (96 %, an increase of 16% compared to pre–VARK results), compared to 23 participants in the TT cohort (92%, the same increase as MR cohort). Additionally, Table 5 presented the frequency of positive signs for the four modalities between the pre–VARK and post–VARK among cohorts. Overall, the Mann–Whitney U test indicates no significant differences in VARK between cohorts in the pre–test and post–test, indicating an approximate response in both groups.

4.5. Analysis of MR cohort's user experience surveys

4.5.1. Training user experience (TUX)

The TUX survey includes statements modified from the workplace learning assessment framework [117] on training objectives, interactions, content, and facilities. These constructs are borrowed and modified from the previous research [35,118–120]. The purpose of the TUX study is to investigate users' subjective evaluations of the MR RPP training platform, namely their psychometric perceptions and intent to utilize this training system. There are 13 criteria studied, which contain several sub–questions (SQs), including perceived utility (PU), perceived ease–of–use (PEOU), perceived enjoyment (PE), perceived behavioral control (PBC), perceived internal control (PIC), attitude (ATT), satisfaction (SAT), confirmation (CON), engagement (ENG), user presence (PRE), user attention (ATTEN).

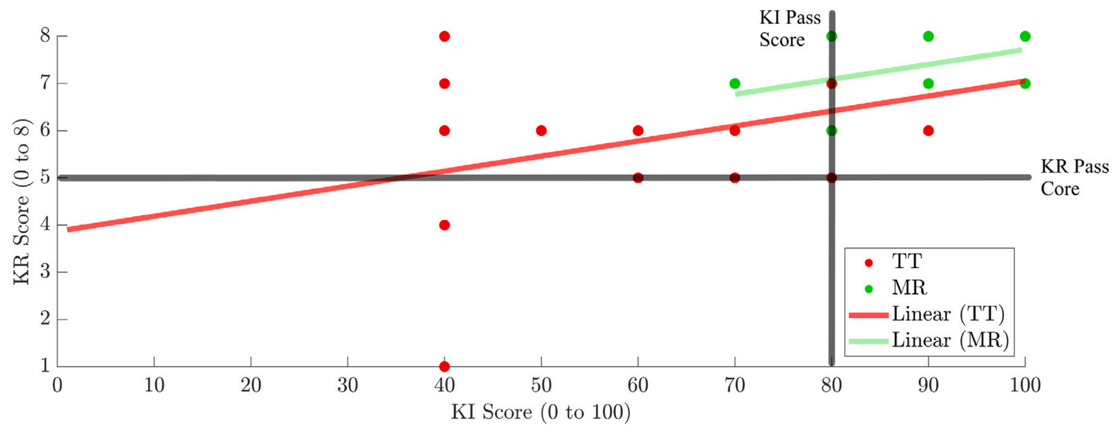

Fig. 7. Correlation between knowledge retention (KR) and knowledge interpretation (KI) scores on both cohorts.

Table 4
Mann–Whitney U test for RPP operations competence between cohorts in human–centric training assessment, NASA task load index & training engagement.

Variable	MR Cohort Mean (SD)	TT Cohort Mean (SD)	Mann–Whitney U	Z	Asymp. Sig. (right–tailed)	Result (Ho or H1)
KR	07.48 (00.65)	06.20 (01.50)	387.50	2.57	0.0051[a]	H1, MR>TT[a]
KI	92.40 (09.70)	73.20 (22.12)	475.00	3.96	0.0000[a]	H1, MR>TT[a]
NASA–TLX	48.41 (12.47)	47.87 (13.89)	323.50	0.20	0.4192	Ho, MR≤TT
MD	38.40 (20.14)	45.60 (24.72)	256.00	−1.12	0.8681	Ho, MR≤TT
PD	18.20 (17.07)	15.40 (16.58)	373.50	1.21	0.1131	Ho, MR≤TT
TD	37.60 (26.50)	39.80 (29.56)	312.00	−0.02	0.5078	Ho, MR≤TT
P	89.00 (09.46)	72.40 (22.51)	453.50	2.76	0.0031[a]	H1, MR>TT[a]
E	46.00 (30.28)	49.00 (23.05)	287.00	−0.51	0.6941	Ho, MR≤TT
F	27.00 (19.79)	29.40 (25.26)	314.00	0.02	0.4922	Ho, MR≤TT
TE	03.96 (00.79)	03.16 (00.62)	474.00	3.37	0.0003[a]	H1, MR>TT[a]

[a] $\rho < 0.05 \rightarrow$ Ho is rejected.

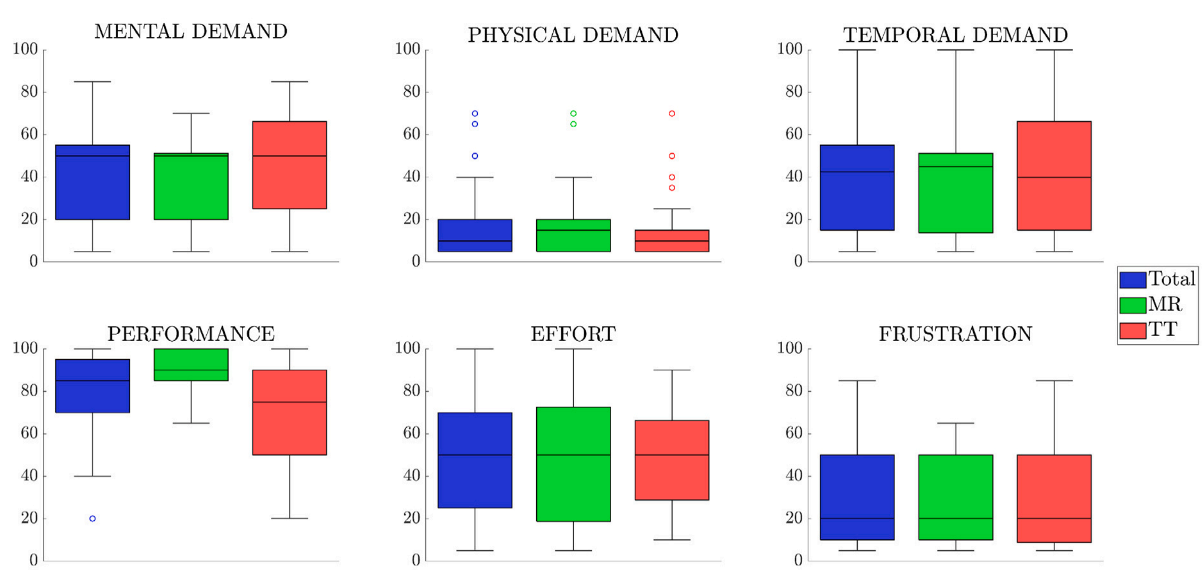

Fig. 8. Boxplots of individual NASA Task Load Index (NASA–TLX) subscale scores.

Table 5
The list of changes in VARK modalities between the Pre–VARK & Post–VARK in both cohorts.

Modality	MR cohort (n = 25)		TT cohort (n = 25)	
	Pre–VARK n (%)	Post–VARK n (%)	Pre–VARK n (%)	Post–VARK n (%)
Vs	1 (4%)	0 (0%)	2 (8%)	2 (8%)
As	1 (4%)	1 (4%)	3 (12%)	0 (0%)
Rs	3 (12%)	0 (0%)	1 (4%)	0 (0%)
Ks	20 (80%	24 (96%)	19 (76%)	23 (92%)

Table 6
Wilcoxon's signed rank test for VARK on the cohorts towards two training conditions.

Modality	MR cohort (n = 25)		TT cohort (n = 25)	
	Z	Asymp. Sig. (two–tailed)	Z	Asymp. Sig. (two–tailed)
Vs	1.560	0.119	0.733	0.464
As	−0.631[a]	0.528	−0.194[a]	0.846
Rs	−0.738[a]	0.460	−0.589[a]	0.556
Ks	2.343	0.019[b]	2.016	0.044[b]

[a] According to negative sign.
[b] $\rho < 0.05$.

The last two criteria, behavioral intention to use the proposed MR system/MR technology for training and learning (BIUMRS/BIUMRT), are further derived to provide the participants' perception of utilizing the proposed platform and the MR technology.

Since each TUX criteria contains multi–cross–check–questions, Cronbach's Alpha test is invoked to measure the internal consistency of the questionnaires on the TUX survey to ensure the reliability of the collected dataset (ranges between 0 and 1) [121]. The PEOU, PIC, PRE, and BIUMRT criteria have coefficients of Cronbach's alpha between 0.7 – 0.8, which indicates acceptable internal consistencies. There is only the perceived behavioral control (PBC) criterion has a questionable internal consistency (α = 0.679). The rest of the criteria have Cronbach's alpha above 0.8, which is considered a good internal consistency. Some criteria have a coefficient that nearly reaches 0.9 (such as CON and ATTEN), which is excellent internal consistency [122] as in Table 7.

Regarding TUX results, most of the criteria are perceived at an agreement level (mean is around 4.0) by the MR cohort, except PIC and PRE criteria. The agreement level on each criterion means that participants give positive support and perception to the MR training platform in the corresponding criterion. However, there is a consideration between the "Neutral" and "Agree" levels in the PIC criterion (mean ≈ 3.5). The PRE mean is around 3.3 which has acceptable coefficients ($\alpha \approx 0.74$), hence is no discussion on this criterion. Nevertheless, the overall Cronbach's alpha of the TUX survey is 0.933, which highlights the "Excellent" internal consistencies of this dataset (excluding 7 reversed SQs in ENG and ATTEN criteria).

4.5.2. SDS, SCLS, and EPQ

A total of 20 SDS sub–questions are used to assess participants' perspectives on MR training objectives, information, problem–solving, and fidelity. It was designed to evaluate the five design features of the instructor–developed simulations in MR training. SCLS is a 13–item instrument designed to measure student satisfaction with simulation activities and self–confidence in learning. EPQ subscale includes 16 sub–questions that evaluate active learning and training participation opportunities. SCLS and EPQ subscales are evaluated by summing the replies to the Likert statements, which ranged from 1 point for strongly disagree to 5 points for strongly agree. Higher SCLS score is linked to higher levels of happiness and self–confidence. Higher EPQ scores indicate a more comprehensive awareness of educational practices.

The Cronbach's alpha test also is employed to examine the reliability of the SDS, SCLS, and EPQ datasets. In Table 8, it is observed that the coefficients of their Cronbach's alpha are in the acceptable range (0.8 > $\alpha \geq$ 0.7). The mean values of these three are around 4.0, which is the "Agree" level. The standard deviations are around 0.8, except SCLS (SD = 0.697). Overall, there is no significant difference in perspective for the remaining statements in the SCLS, SDS, and EPQ survey outcomes. Thus, these results reported the satisfaction, and self–confidence in the knowledge achievement, and the efficiency of MR digital object contents based on participants' perspectives.

5. Discussions

5.1. Principal findings

5.1.1. RQ1 & H1 – user knowledge retention & interpretation

The RQ1 asked about the differences in perceived user knowledge retention and interpretation relative to the training conditions. Both methods bring an acceptable pass rate on the KR assessment, despite perceptions of the learning environment differences among the two cohorts. However, the results of the quantitative analysis strongly suggest that the MR cohorts maintain a significantly higher pass rate than the TT cohort in the practical KI assessment.

The reason for this is that the MR participants had opportunities to learn from directive interaction and collaboration aiding with the virtual model and digital twin cobot cell in 360–ATE, while TT participants needed to attend to the theories and practice separately in order as usual. As a result, the participants in the TT cohort responded that they had gained knowledge, but they were not confident that learning actively occurred. As reflected by some participants via the TE survey, the traditional training lacked an entertainment factor, estimating why engagement and associated learning were not encouraged in the cohort. On the other hand, the MR cohort enjoyed and were satisfied with the experience. Human–centric collaborative training resonating with the digital twin does have its positive impacts, matching with the situated learning theory which experiential training and direct interaction with the object of instruction prove beneficial for the trainees.

Nevertheless, the H1 posits that knowledge retention and interpretation of the two cohorts are the same. Although the instruction effectively increases measured knowledge about robotic technical in both cohorts, which shows that both training conditions are beneficial for instructional purposes in the KR, participants in the MR show a significantly higher knowledge interpretation with physical cobot. They completed the training with higher KI scores than participants in the TT cohort (92% passed the test), even though the MR cohort must quickly adapt to the new training platform as Fig. 7. The knowledge retention and interpretation results of the TT cohort contradict the initial perception of usual familiar training such as traditional classroom style. These findings present with the exception that collaborative MR training is constantly delivering better instructional results than TT. The exception is the personal innovativeness level in immersive technologies where the high PIL leads to higher adaptability in the MR cohort and maintains the gain compared to the TT cohort for the KI and KR assessment.

The rejection of this hypothesis might be explained by both the user–friendly convenience of the collaborative mixed reality technology, combined with digital twin and kinesthetic learning. As mentioned, the analysis results against this assumption, with many MR participants successfully passing the final assessment with the top scores. The TT cohort is affected by the traditional training which reduces their attention, and engagement leads to the loss of new knowledge. The results for the RQ1 support previous research findings that showed an increased engagement with differences in retention gain when comparing knowledge retention or score gains [123,124]. For instance, an experimental study exploring audio processing in mixed reality to boost motivation in piano learning [125] found that curiosity

Table 7
Cronbach's alpha of TUX subscales, mean & standard deviation in SQs/reversed SQs.

No.	Criteria	# of SQs	Mean/Mean[a]	SD/SD[a]	α/α[a]	Reliability
1	PU	3	4.253/–	0.680/–	0.856/–	Good
2	PEOU	4	4.360/–	0.482/–	0.723/–	Acceptable
3	PE	3	4.027/–	0.636/–	0.860/–	Good
4	PBC	3	4.187/–	0.392/–	0.679/–	Questionable
5	PIC	10	3.536/–	1.049/–	0.751/–	Acceptable
6	ATT	3	4.133/–	0.723/–	0.806/–	Good
7	SAT	4	4.160/–	0.587/–	0.827/–	Good
8	CON	3	4.040/–	0.743/–	0.896/–	Good
9	ENG	10 (4[a])	3.973/2.410[a]	0.714/0.726[a]	0.809/0.815[a]	Good
10	PRE	3	3.333/–	0.977/–	0.742/–	Acceptable
11	ATTEN	10 (3[a])	4.297/2.360[a]	0.458/0.864[a]	0.880/0.869[a]	Good
12	BIUMRS	3	4.093/–	0.597/–	0.865/–	Good
13	BIUMRT	3	4.000/–	0.885/–	0.754/–	Acceptable

[a] Reversed.

Table 8
Summary of mean, standard deviation & Cronbach's alpha of TUX, SDS, SCLS & EPQ.

Survey	Mean	SD	α	Reliability
TUX (62 SQs)	3.979[a]	0.814[a]	0.933[a]	Excellent
SDS (20 SQs)	4.024	0.813	0.713	Good
SCLS (13 SQs)	4.037	0.697	0.777	Good
EPQ (16 SQs)	3.952	0.864	0.742	Good

[a] Excluding 7 reversed SQs.

and joy were the most significant factors influencing the use of the application. The effectiveness of the learning was moderate, with an increase of 31.28% and it is utilized and improved further to keep users motivated.

5.1.2. RQ2/3 & H2/3 – user engagement, attention & knowledge delivery

The RQ2 and RQ3 examine the differences in user engagement, attention, and their perceived digital twin in 360–ATE as a training tool. The quantitative data suggested a positive impact of human–centric collaborative MR training by participants, almost all being excited from interacting with the digital twin cobot, which provided a unique opportunity to learn and practice in a blended environment. Although both groups were exposed to the same training material, theories, and assessment, there was less enthusiasm about the TT cohort's content and engagement in training. Quantitative data support this valuation, indicating that the MR participants (Mean–TE = 3.96/5.00, SD = 0.79, Agree) had a higher opinion about the information presented than the TT participants (Mean–TE = 3.16/5.00, SD = 0.62, Neutral). Nevertheless, some users in the TT cohort reflected that the classroom style was classical, and familiar, helping them easily follow without concerns as the SD–TE of the TT cohort is less than MR one. This observation is consistent with the instructional impact proposed by the situated learning theory. Interacting with a digital twin cobot cell anchored at an allocated surface, much as a presence at the physical station, even with more safety protocol helped with engagement and retention.

The H2 assumes that MR participants will demonstrate higher levels of engagement with digital content than TT participants in the TT cohort. The hypothesis is fully supported by the static results in Table 4 (TE) and Table 7 (ATT, SAT, and ENG), with participants in the MR cohort reporting higher levels of engagement with the learning environment than those in the TT cohort. In addition, the hypothesis of RQ3 assumes that the MR cohort precepts that digital twin cobot enhances and aids the simulation training process. The outcomes of the TUX analysis strongly support the H3, except for the perceived behavioral control subscale. The Croroach's alpha test within PBC is 0.679 less than an acceptable range (0.7) leading to questionable internal consistency. This happens due to the free–style training in the new mixed environment which requires high adaptability.

Given the fixed training of the TT cohort and the dynamic training of the MR cohort, the unit technical KI assessment was chosen to measure the proportion of the training performance that involves physical cobot operations. In a recent position paper, meta–researchers argued that standardized techniques of assessment for mixed reality are unfeasible and undesirable. The approach [126,127] taken by the authors in developing certain metrics for MR interaction analysis is in line with their suggestions. Rather than increased standardization, a greater diversity of goal–specific measuring equipment is being considered for the future. As a proportion of overall performance, H2 and H3 are found to be in support. This can happen for various reasons, including the absence of experimental fatigue and stress, when interacting with the digital twin, MR participants do not deal with safety and human errors as on physical robot. Moreover, the MR cohort is not coerced into training (they chose the training context and controlled the robot themselves), which increases user active learning skills [128].

5.1.3. RQ4/5 & H4/5 – user effort & system intentions utilization

The RQ4 investigates what difference in the effort demand between two cohorts in two training conditions. According to the NASA–TLX assessment, both cohorts perceived the medium attention in effort demand as Table 4 around 50/100. The right–tailed Mann–Whitney U test indicates that the MR cohort takes less effort than or equal to the TT cohort. This result leads to the acceptance of H4 that the MR cohort will spend less effort than the TT cohort.

Although participants in the MR cohort started their first experience with MR–based training instruction, it is estimated that the human–centric factor had a positive effect. All MR participants found the 360–ATE beneficial for instruction and considered it a human–centric application not limited to topics that could benefit from high visual engagement, like collaborative training. These findings are consistent with previous research on assisted learning using ImT. Furthermore, progressive exposure of learners to mixed environments may produce habituation and decrease overall effort with the medium, allowing novice learners to acquire new knowledge.

The RQ5 asks if there is an enhancement in competence and augmented cognitive capabilities among students and trainees in automation, which extends to related training fields, with the incorporation of human–centric MR technology for human–robot collaboration training on robot pick–and–place operations. Initially, the H5 assumed that using the proposed MR platform is a concern due to its technology adaptability and knowledge requirement. As the findings above, it is reported the positive acceptance of using this MR training and future perception to use these collaborative MR–based training systems (Mean–BIUMRS = 4.093/5.0 and Mean–BIUMRT = 4.0/5.0 as Table 7). In addition, all remaining surveys such as SDS, SCLS, and EPQ also against this hypothesis since they indicate positive responses with the MR training platform.

5.1.4. Summary findings

Some scholars argue that comparative studies are out of date and that scientists should focus on the learning processes that students go through when immersive technologies are employed in the classroom [129,130]. Everyone acknowledges the need to know when and how to use mixed reality in the classroom. Nonetheless, considering the results of this investigation, it is equally critical to understand how users engage with mixed reality technology, especially in light of the significance of integrating digital objects and models into the educational process (a primary benefit of employing immersive technology).

Because the 360–ATE allows for a more engaging learning environment than a traditional classroom setting where students may sit through lectures or passively follow the movements of a pointer on a screen, this study's results suggest that understanding how MR's interactive features can improve knowledge retention is crucial for implementing MR in the classroom. Based on these findings, educational designers, teachers who are open to integrating MR, and other interested parties should schedule more time for training so that students can become proficient in the new setting, improving instruction and assisting with the industrialization of the digital transition.

5.2. Limitations and future work

While this study offers valuable insights, it is crucial to acknowledge limitations. Initially, the participant sample size was inherently small due to the research being conducted within a university and industry partner curriculum. This limited sample size, added with a relatively young age range (20–28) as Section 4.1, may constrain the generalizability of the findings. To augment the research's validity and reliability, future studies could consider incorporating a larger and more demographically diverse sample across age groups.

Moreover, the study focused on a specific group of participants, including students and novice trainees from particular educational institutions and industrial partners, as mentioned in Section 3.4. This focus may limit the generalizability of the findings to a broader population. Future research should include participants from diverse educational settings, personal innovativeness levels, and backgrounds, such as automation, software, IT, and other related fields, to ensure the findings are more representative and applicable to a transdisciplinary context.

In addition, the study was conducted within a specific robotic technical context, which may limit the transferability of the findings to other industrial settings. Future research needs to investigate the impact of MR and ImT in various industrial contexts to gain a more comprehensive understanding of how human–centric immersive technologies, especially MR, can be utilized for industrial training and advanced skill development across diverse populations.

In terms of technological limitations, it is important to recognize that the effectiveness of MR for technical training may be influenced by the specific MR and AR system used in the study. Different immersive technologies or platforms may yield different results, and it is important to acknowledge that the findings may not cover other MR applications or tools. Future research could explore the impact of various MR systems and platforms to understand their effectiveness on collaborative training, human resource development, and human–centric design for transdisciplinary applications.

6. Conclusions

Mixed reality training has been expected to provide various advantages in the industrial domain through industry expert interviews and extended reviews of commercial applications. In light of the discussion above, we present the impact of the human–centric MR training platform in providing immersive human–robot interaction training in the immersive 360° Digital–Objects–Based Augmented Training Environment, partnering with ABB Robotics Vietnam for RPP training with a digital twin ABB GoFa™ CRB–15000 cobot station, and the user training assessment. The training approach embeds a kinesthetic learning strategy into collaborative robot pick–and–place training in the immersive environment where the users are allowed to interact, control, configure, and simulate the robot program with its digital twin before loading the final program on the physical robot station.

Training strategies are tested on two groups of participants: industrial novice trainees and university students. The study compared the MR training platform to the traditional classroom training method. It has been discovered that MR training is both engaging and effective in developing learner confidence, particularly for beginning users by performing a similar empirical comparison of traditional and mixed reality training on novices. The findings contribute to the industrial training domain in combination with active kinesthetic learning as a reflection for program course coordinators, immersive technologies developers, researchers, educators, and stakeholders in industrial settings and related sectors. These results also aid in the digital transition of modern companies towards a transdisciplinary and human–centric system model in terms of training and human resource development.

The self–reflection data collected from user experience of two training conditions has proved and verified the application of mixed reality technology as metaphors for the mentioned industrial training via several questionnaires and tests, including Pre/Post Visual, Aural, Read/Write, Kinesthetic – VARK questionnaires, Training Engagemen – TE, Post–Training Tests (Knowledge Interpretation – KI and Knowledge Retention – KR), NASA Task Load Index – NASA–TLX assessment, Training User Experience – TUX questionnaire, Satisfaction and Self–Confidence in Learning Scale – SCLS, Simulation Design Scale – SDS, and Educational Practices Questionnaire – EPQ. Several statistical techniques were invoked for data analysis, such as Spearman's Rank Correlation, Mann–Whitney U Test, Wilcoxon's Signed Rank Test, and Cronbach's Alpha to validate and analyze the dataset.

The study found that MR participants effectively boost their safety, confidence, and performance, augmenting cognitive capabilities. The virtual training content of the collaborative MR training grabs the participant's attention and stimulates their active learning abilities. Thus, participants perceived the usefulness of the proposed training platform which predicted the future intention to use this training approach and other similar 360–ATE MR–assisted training systems. There is no significant difference in the participant's effort spent in each training condition reporting at the medium level.

Overall, this work outlines that the human–centric collaborative MR training platform with digital twin cobot and kinesthetic learning in 360–ATE outweighs the conventional training method in terms of user engagement and technical performance. According to the findings, it can serve as an alternative approach for novice students and industrial trainees in robot pick–and–place operations. However, there are still concerns and research gaps that require further investigation to ascertain whether the proposed mixed reality training approach and digital twins technology can achieve widespread adoption, be universally applicable, and significantly advance transdisciplinary applications and industry growth in the digitization and cloud migration eras.

CRediT authorship contribution statement

Thien Tran: Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Quang Nguyen:** Writing – original draft, Visualization, Investigation, Formal analysis. **Toan Luu:** Writing – original draft, Software, Formal analysis. **Minh Tran:** Writing – review & editing, Validation, Supervision, Resources, Investigation, Funding acquisition, Data curation, Conceptualization. **Jonathan Kua:** Conceptualization, Supervision, Validation, Writing – review & editing. **Thuong Hoang:** Writing – review & editing, Validation, Supervision, Conceptualization. **Man Dien:** Validation, Software, Resources, Funding acquisition.

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Ethics approval

This research study, with project number **#26910**, has been approved by the RMIT University CHEAN (College Human Ethics Advisory Network) as it meets the requirements of the National Statement on Ethical Conduct in Human Research (NHMRC, 2007).

Consent to participate

An informed consent was obtained from all individual participants included in the study, explaining what their participation entailed as well as the voluntary nature of the study and the possibility to withdraw from the study at any time without reason.

Consent for publication

Consent to publish was obtained from all the partners involved in the project. Furthermore, data were anonymized, and no results were reported on the level of an individual participant.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

See Tables 9 and 10.

Table 9
List of acronyms.

360–ATE	Immersive 360° Digital–Objects–Based Augmented Training Environment
A(s)	Aural/Auditory (oral lessons and conversing)
AR	Augmented Reality
ATT	Attitude
ATTEN	User Attention
BIUMRS	Behavioral intention to use the proposed MR training system
BIUMRT	Behavioral intention to use MR technology for training
Cobot	Collaborative Robot
CON	Confirmation
DT	Digital Twin(s)
E	Effort
ENG	Engagement
EPQ	Educational Practices Questionnaire
F	Frustration
H(s)	Hypothesis/Hypotheses
HRC/HRI	Human–Robot Collaboration/Interaction
HSE/SA	Health–Safety–Environment–Security and Sustainability
ImT	Immersive technologies
K(s)	Kinesthetic (learning by doing)
KI/KR	Knowledge Interpretation/Retention
MD	Mental Demand
MR	Mixed Reality
NASA–TLX	NASA Task Load Index
P	Performance
PBC	Perceived Behavioral Control
PD	Physical Demand
PE	Perceived Enjoyment
PEOU	Perceived Ease–of–Use
PIC	Perceived Internal Control
PIL	Personal Innovativeness Level
PRE	User Presence
PU	Perceived Utility
R(s)	Read/Write (written material)
RPP	Robot Pick–and–Place
RQ(s)	Research Question(s)
SAT	Satisfaction
SCLS	Satisfaction and Self–Confidence in Learning Scale
SD	Standard Deviation
SDS	Simulation Design Scale
SQ(s)	Sub–Question(s)
TD	Temporal Demand
TE	Training Engagement
TT	Traditional Training
TUX	Training User Experience
UI	User Interface
V(s)	Visual (charts, graphs, or symbolic devices)
VARK	Visual, Aural, Read/Write, Kinesthetic

Table 10
Hardware specifications for proposed MR training system.

Device	Chipset/Controller	OS/Platform/Firmware	Specification
Microsoft HoloLens 2 (UI devices)	Qualcomm Snapdragon 850	OS: Windows Holographic Platform: Windows Mixed Reality (UWP)	CPU: Octa–core Kyro 385 (4 × 2.96 GHz, 4 × 1.8 GHz) GPU/HPU: Adreno 630 RAM: 4 GB ROM: 64 GB WiFi: 5 (802.11ac 2 × 2) Bluetooth: 5.0 Tracking type: 6 DoF inside–out via 4 integrated cameras Refresh rate: 60 Hz Resolution: 1440 × 936 per eye Visible FoV: 43° horizontal; 29° vertical; 52° diagonal
Microsoft HoloLens (1) (UI devices)	Intel Atom x5–Z8100	OS: Windows Holographic Platform: Windows Mixed Reality (UWP)	CPU: Quad–core (4 x 1.84 GHz) GPU/HPU: Intel HD Graphics (Cherry Trail) RAM: 2 GB ROM: 64 GB WiFi: 5 (802.11ac) Bluetooth: 4.1 LE Tracking type: 6 DoF inside–out via 4 integrated cameras Refresh rate: 60 Hz Resolution: 1268 × 720 per eye Visible FoV: 30° horizontal; 17° vertical
ABB GoFa™ Cobot CRB–15000 (Cobot Station)	ABB OmniCore™ C30 Controller	Robot OS ABB RobotWare 7.5.1 ABB RobotStudio 2022–2023	Robot type: Collaborative – Cobot Cobot Reach: 950mm (wrist), 1050mm (flange) Payload: 5kg Protection standard: IP54, ISO cleanroom 4 Number of axes: 6 Software special options required: • 3020–1 PROFINET Controller • 3020–2 PROFINET Device • 3023–2 PROFisafe Device • 3024–2 EtherNet/IP Adapter • 3043–3 SafeMove Collaborative • 3114–1 Multitasking • 3116–1 FTP and SFTP client • 3119–1 RobotStudio Connect • 3120–2 FlexPendant Essential App Package • 3121–1 RW Add–in Prepared • 3124–1 Externally Guided Motion (EGM) • 3134–1 101 Data Gateway
Industrial Compact DIN–rail Mounted Cellular Router Hitachi Energy – TRO610 (Internet Router)	Single core 600 MHz 32–bit ARM–processor	NA	Model: DIN–Rail mounted (Cat–4, NB–IoT, Cat–M1 with 2G/3G/4G) Security: X.509 PKI, IKEv2 IPsec VPN, AES–128/256, SHA–384/512 Monitoring: Supros, Network Element Manager, SNMP, Syslog SIM switch: Dual–SIM configuration Routing and IP services: • DHCP, NTP, SNTP server (IPv4, IPv6) • Port Forwarding Ethernet services: • Untagged and 802.1q VLAN tagged (access and trunk modes) • Maximum of 32 VLANs Serial protocols: • Raw UDP/TCP–IP transport • Modbus (RTU, ASCII), DNP3, PG&E2179 and Mirrored Bits • IEC 101 / IEC 104 conversion
Alienware X17 (Central Broker)	Intel Core	Windows 10	CPU: Intel Core i7 – 11800H (8 × 16) Turbo frequency: 4.6 GHz Integrated GPU: Intel UHD Graphics for 11th Gen Dedicated GPU: GeForce RTX 3060 6 GB RAM: 64 GB – DDR4 3200 MHz WIFI: v6 Bluetooth: 5.2 NIC: Killer E3100G

Data availability

The data that has been used is confidential.

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